MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, RECEIVING

## TYPE 1AE4

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, sharp cutoff

| | | |
|---|---|---|
| Outline | --- | 5-2 (EIA) |
| Base | --- | E7-1 |
| Envelope | --- | T5-1/2 |
| Cathode | --- | Coated filament |

Base connections:

| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| Element | --- | -f, g3 sd | a | g2 | nc | -f, g3 sd | g1 | +f |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Ec1 | Eb | Ec2 | Ik | Alt |
|---|---|---|---|---|---|---|
| Unit: | Vdc | Vdc | Vdc | Vdc | mAdc | ft |
| Maximum: | 1.50 | --- | 100 | 100 | 11 | (see note 1) |
| Minimum: | 1.00 | --- | --- | --- | --- | |
| TEST CONDITIONS: | 1.25 | 0 | 90 | 90 | --- | --- |

*GENERAL:*

Qualification - Required

(B) denotes changes



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | | |
| 1266 | Total grid current | 2 | Ec1 = -0.5 Vdc | 1.0 | II | Ic1 | --- | -0.5 | μAdc |
| 1306 | Transconductance (2) | 2 | Ef = 1.0 Vdc | 1.0 | II | Sm | 950 | 2,000 | μmhos |
| 1201 | Tap shorts | - | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | | |
| 1031 | Low-frequency vibration | - | Rp = 10,000 ohms | --- | --- | Ep | --- | 500 | mVac |
| 1301 | Filament current | - | | --- | --- | If | 88 | 112 | mAdc |
| 1256 | Electrode current (1) (anode) | - | | --- | --- | Ib | 2.0 | 5.0 | mAdc |
| 1256 | Electrode current (2) (anode) | - | Ec1 = -5 Vdc | --- | --- | Ib | --- | 50 | μAdc |
| 1256 | Electrode current (screen) | - | | --- | --- | Ic2 | 0.5 | 1.5 | mAdc |
| 1311 | Anode resistance | - | | --- | --- | rp | 0.25 | --- | Meg |
| 1306 | Transconductance (1) | - | | --- | --- | Sm | 1,100 | 2,000 | μmhos |
| 1331 | Direct-interelectrode capacitance | - | Shield No. 316 | --- | --- | Cg1p<br>Cin<br>Cout | ---<br>2.8<br>3.0 | 0.008<br>4.4<br>5.8 | pF<br>pF<br>pF |
| (B) 1121 | Base strain | - | | --- | --- | --- | --- | --- | --- |
| (B) 2126 | Glass strain | - | | --- | --- | --- | --- | --- | --- |
| (B) 1105 | Permanence of marking | - | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | | |
| 1501 | Intermittent life | - | Group A; Ef = 1.25 Vdc or Vac with equivalent Ec | --- | --- | --- | --- | --- | --- |
| --- | Intermittent life-test end point (500 hours): | | | | | | | | |
| 1306 | Transconductance (1) | - | | --- | --- | Sm | 950 | --- | μmhos |

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.



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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2596)



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